

MINISTRY OF EDUCATION AND HIGHER EDUCATION

FORM FOUR EXAMS, 2019

PHYSICS



P/LAND NATIONAL EXAMINATION BOARD

MINISTRY OF EDUCATION AND HIGHER EDUCATION
PUNTLAND NATIONAL EXAMINATIONS BOARD

Code Number

FORM FOUR EXAMINATION 2019
TIME 2 HOURS AND 10 MINUTES FOR READING

PHYSICS

Instructions to candidates

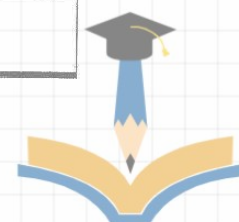
- Answer all the questions
- This paper consists of 15 pages, count it and if any is missing inform your invigilator
- Do not write your **name and roll number** on the exam paper
- Make sure that **student's profile** is attached to the exam paper, if not, inform you invigilator.
- No extra paper is allowed. Rough work can be done on page 1, 14 and 15. those will not be marked.
- If you make a mistake, **cross out the incorrect** answer and **write your correct answer**.

This exam paper consists of following parts

Parts	Marks
Part one: Multiple Choice	10 marks
Part two: Structured Questions	90 marks
Total: 100 Marks	

For the markers only

PARTS	MARKS
Part one	
Part two	
TOTAL	%

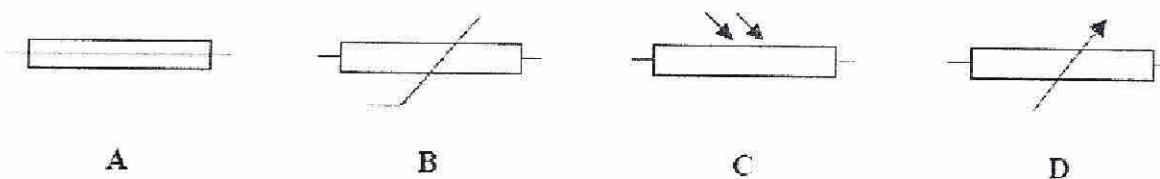


1. $\frac{1}{2} \times \frac{3}{4} = \frac{1 \times 3}{2 \times 4} = \frac{3}{8}$
 2. $\frac{2}{3} \times \frac{5}{6} = \frac{2 \times 5}{3 \times 6} = \frac{10}{18} = \frac{5}{9}$
 3. $\frac{3}{5} \times \frac{4}{7} = \frac{3 \times 4}{5 \times 7} = \frac{12}{35}$
 4. $\frac{1}{4} \times \frac{2}{3} = \frac{1 \times 2}{4 \times 3} = \frac{2}{12} = \frac{1}{6}$
 5. $\frac{5}{8} \times \frac{3}{10} = \frac{5 \times 3}{8 \times 10} = \frac{15}{80} = \frac{3}{16}$
 6. $\frac{7}{9} \times \frac{2}{5} = \frac{7 \times 2}{9 \times 5} = \frac{14}{45}$
 7. $\frac{4}{6} \times \frac{1}{2} = \frac{4 \times 1}{6 \times 2} = \frac{4}{12} = \frac{1}{3}$
 8. $\frac{6}{8} \times \frac{3}{4} = \frac{6 \times 3}{8 \times 4} = \frac{18}{32} = \frac{9}{16}$
 9. $\frac{2}{5} \times \frac{7}{10} = \frac{2 \times 7}{5 \times 10} = \frac{14}{50} = \frac{7}{25}$
 10. $\frac{3}{7} \times \frac{5}{12} = \frac{3 \times 5}{7 \times 12} = \frac{15}{84} = \frac{5}{28}$
 11. $\frac{1}{3} \times \frac{4}{5} = \frac{1 \times 4}{3 \times 5} = \frac{4}{15}$
 12. $\frac{5}{6} \times \frac{2}{7} = \frac{5 \times 2}{6 \times 7} = \frac{10}{42} = \frac{5}{21}$
 13. $\frac{3}{8} \times \frac{6}{9} = \frac{3 \times 6}{8 \times 9} = \frac{18}{72} = \frac{1}{4}$
 14. $\frac{7}{10} \times \frac{1}{3} = \frac{7 \times 1}{10 \times 3} = \frac{7}{30}$
 15. $\frac{4}{5} \times \frac{3}{7} = \frac{4 \times 3}{5 \times 7} = \frac{12}{35}$
 16. $\frac{2}{9} \times \frac{5}{11} = \frac{2 \times 5}{9 \times 11} = \frac{10}{99}$
 17. $\frac{6}{7} \times \frac{1}{4} = \frac{6 \times 1}{7 \times 4} = \frac{6}{28} = \frac{3}{14}$
 18. $\frac{8}{10} \times \frac{3}{5} = \frac{8 \times 3}{10 \times 5} = \frac{24}{50} = \frac{12}{25}$
 19. $\frac{1}{5} \times \frac{6}{8} = \frac{1 \times 6}{5 \times 8} = \frac{6}{40} = \frac{3}{20}$
 20. $\frac{5}{7} \times \frac{4}{9} = \frac{5 \times 4}{7 \times 9} = \frac{20}{63}$
 21. $\frac{3}{10} \times \frac{2}{3} = \frac{3 \times 2}{10 \times 3} = \frac{6}{30} = \frac{1}{5}$
 22. $\frac{7}{12} \times \frac{5}{14} = \frac{7 \times 5}{12 \times 14} = \frac{35}{168} = \frac{5}{24}$
 23. $\frac{4}{9} \times \frac{3}{6} = \frac{4 \times 3}{9 \times 6} = \frac{12}{54} = \frac{2}{9}$
 24. $\frac{6}{11} \times \frac{1}{2} = \frac{6 \times 1}{11 \times 2} = \frac{6}{22} = \frac{3}{11}$
 25. $\frac{2}{7} \times \frac{8}{10} = \frac{2 \times 8}{7 \times 10} = \frac{16}{70} = \frac{8}{35}$
 26. $\frac{5}{12} \times \frac{4}{5} = \frac{5 \times 4}{12 \times 5} = \frac{20}{60} = \frac{1}{3}$
 27. $\frac{3}{11} \times \frac{6}{13} = \frac{3 \times 6}{11 \times 13} = \frac{18}{143}$
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 29. $\frac{4}{13} \times \frac{5}{6} = \frac{4 \times 5}{13 \times 6} = \frac{20}{78} = \frac{10}{39}$
 30. $\frac{1}{6} \times \frac{7}{8} = \frac{1 \times 7}{6 \times 8} = \frac{7}{48}$
 31. $\frac{5}{9} \times \frac{3}{10} = \frac{5 \times 3}{9 \times 10} = \frac{15}{90} = \frac{1}{6}$
 32. $\frac{2}{11} \times \frac{4}{7} = \frac{2 \times 4}{11 \times 7} = \frac{8}{77}$
 33. $\frac{6}{14} \times \frac{1}{3} = \frac{6 \times 1}{14 \times 3} = \frac{6}{42} = \frac{1}{7}$
 34. $\frac{8}{16} \times \frac{3}{5} = \frac{8 \times 3}{16 \times 5} = \frac{24}{80} = \frac{3}{10}$
 35. $\frac{1}{8} \times \frac{5}{9} = \frac{1 \times 5}{8 \times 9} = \frac{5}{72}$
 36. $\frac{5}{14} \times \frac{2}{3} = \frac{5 \times 2}{14 \times 3} = \frac{10}{42} = \frac{5}{21}$
 37. $\frac{3}{16} \times \frac{4}{5} = \frac{3 \times 4}{16 \times 5} = \frac{12}{80} = \frac{3}{20}$
 38. $\frac{7}{18} \times \frac{1}{2} = \frac{7 \times 1}{18 \times 2} = \frac{7}{36}$
 39. $\frac{4}{21} \times \frac{3}{4} = \frac{4 \times 3}{21 \times 4} = \frac{12}{84} = \frac{1}{7}$
 40. $\frac{6}{25} \times \frac{5}{6} = \frac{6 \times 5}{25 \times 6} = \frac{30}{150} = \frac{1}{5}$
 41. $\frac{2}{17} \times \frac{8}{9} = \frac{2 \times 8}{17 \times 9} = \frac{16}{153}$
 42. $\frac{5}{19} \times \frac{3}{4} = \frac{5 \times 3}{19 \times 4} = \frac{15}{76}$
 43. $\frac{3}{20} \times \frac{6}{7} = \frac{3 \times 6}{20 \times 7} = \frac{18}{140} = \frac{9}{70}$
 44. $\frac{7}{22} \times \frac{1}{3} = \frac{7 \times 1}{22 \times 3} = \frac{7}{66}$
 45. $\frac{4}{23} \times \frac{5}{6} = \frac{4 \times 5}{23 \times 6} = \frac{20}{138} = \frac{10}{69}$
 46. $\frac{1}{24} \times \frac{7}{8} = \frac{1 \times 7}{24 \times 8} = \frac{7}{192}$
 47. $\frac{5}{25} \times \frac{3}{4} = \frac{5 \times 3}{25 \times 4} = \frac{15}{100} = \frac{3}{20}$
 48. $\frac{2}{26} \times \frac{4}{5} = \frac{2 \times 4}{26 \times 5} = \frac{8}{130} = \frac{4}{65}$
 49. $\frac{6}{27} \times \frac{1}{2} = \frac{6 \times 1}{27 \times 2} = \frac{6}{54} = \frac{1}{9}$
 50. $\frac{8}{28} \times \frac{3}{7} = \frac{8 \times 3}{28 \times 7} = \frac{24}{196} = \frac{6}{49}$
 51. $\frac{1}{29} \times \frac{5}{6} = \frac{1 \times 5}{29 \times 6} = \frac{5}{174}$
 52. $\frac{5}{30} \times \frac{2}{3} = \frac{5 \times 2}{30 \times 3} = \frac{10}{90} = \frac{1}{9}$
 53. $\frac{3}{31} \times \frac{4}{5} = \frac{3 \times 4}{31 \times 5} = \frac{12}{155}$
 54. $\frac{7}{32} \times \frac{1}{4} = \frac{7 \times 1}{32 \times 4} = \frac{7}{128}$
 55. $\frac{4}{33} \times \frac{6}{7} = \frac{4 \times 6}{33 \times 7} = \frac{24}{231} = \frac{8}{77}$
 56. $\frac{6}{34} \times \frac{5}{6} = \frac{6 \times 5}{34 \times 6} = \frac{30}{204} = \frac{5}{34}$
 57. $\frac{2}{35} \times \frac{8}{9} = \frac{2 \times 8}{35 \times 9} = \frac{16}{315}$
 58. $\frac{5}{36} \times \frac{3}{4} = \frac{5 \times 3}{36 \times 4} = \frac{15}{144} = \frac{5}{48}$
 59. $\frac{3}{37} \times \frac{4}{5} = \frac{3 \times 4}{37 \times 5} = \frac{12}{185}$
 60. $\frac{7}{38} \times \frac{1}{2} = \frac{7 \times 1}{38 \times 2} = \frac{7}{76}$
 61. $\frac{4}{39} \times \frac{6}{7} = \frac{4 \times 6}{39 \times 7} = \frac{24}{273} = \frac{8}{91}$
 62. $\frac{6}{40} \times \frac{5}{6} = \frac{6 \times 5}{40 \times 6} = \frac{30}{240} = \frac{1}{8}$
 63. $\frac{2}{41} \times \frac{8}{9} = \frac{2 \times 8}{41 \times 9} = \frac{16}{369}$
 64. $\frac{5}{42} \times \frac{3}{4} = \frac{5 \times 3}{42 \times 4} = \frac{15}{168} = \frac{5}{56}$
 65. $\frac{3}{43} \times \frac{4}{5} = \frac{3 \times 4}{43 \times 5} = \frac{12}{215}$
 66. $\frac{7}{44} \times \frac{1}{4} = \frac{7 \times 1}{44 \times 4} = \frac{7}{176}$
 67. $\frac{4}{45} \times \frac{6}{7} = \frac{4 \times 6}{45 \times 7} = \frac{24}{315} = \frac{8}{105}$
 68. $\frac{6}{46} \times \$



PART ONE: MULTIPLE CHOICE QUESTIONS (10 MARKS)

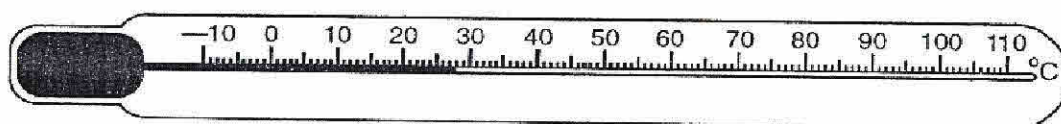
1. Which of these is a symbol for a Thermistor?



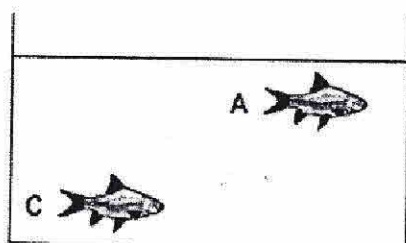
2. Omar Samatar Secondary school students are measuring the room temperature on sunny day using the thermometer below.

The reading of the thermometer is

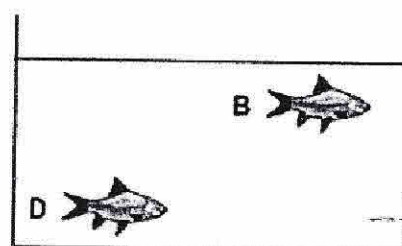
- A. 30°C
B. 20°C
C. 28°C
D. 25°C



3. Two identical fish tanks are filled to the same level with water. One tank contains fresh water and the other tank contains sea-water. Which fish experiences the greatest pressure?

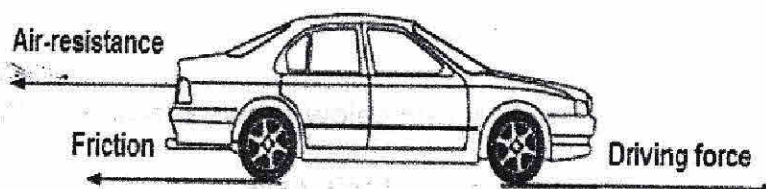


Fresh water



Sea water

4. Three horizontal forces act on a car that is moving a long straight, level road.

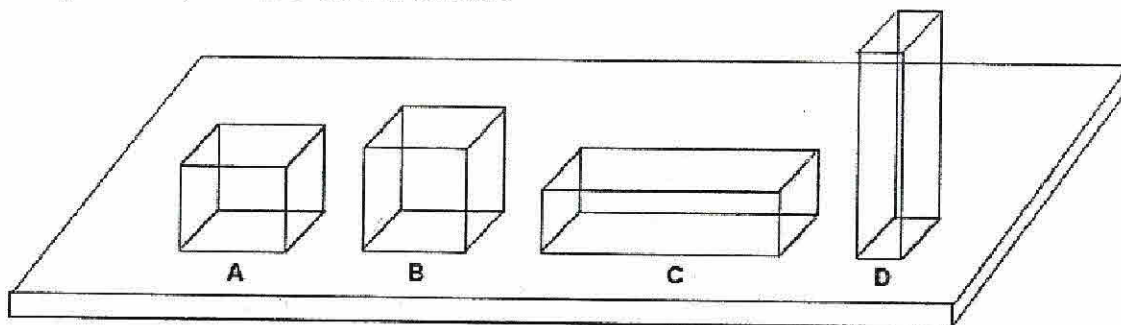


Which combination of the forces would result the car to move at constant speed?

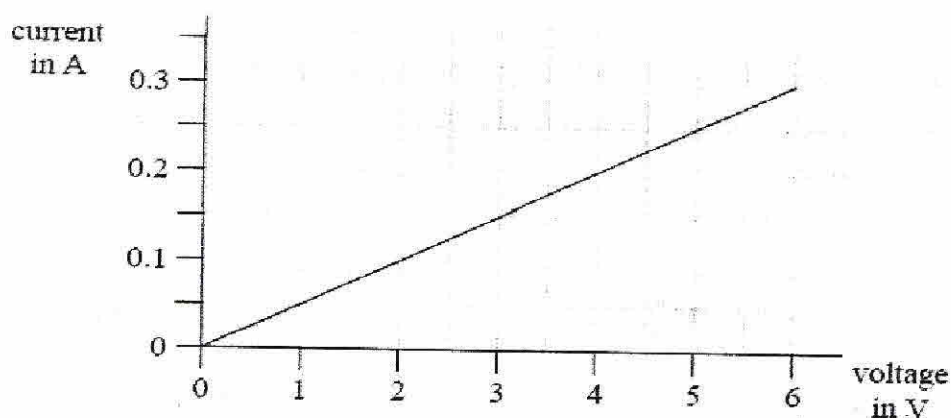
	air resistance	friction	driving force
A	200 N	1000 N	800 N
B	800 N	1000 N	200 N
C	800 N	200 N	1000 N
D	1000 N	200 N	800 N



5. The four objects A, B, C and D all have the same weight. Which one produces the greatest pressure on the bench?



6. Some students in Al-furqan secondary school investigate how the current in a resistor varies with voltage. The students plot a graph of their results.



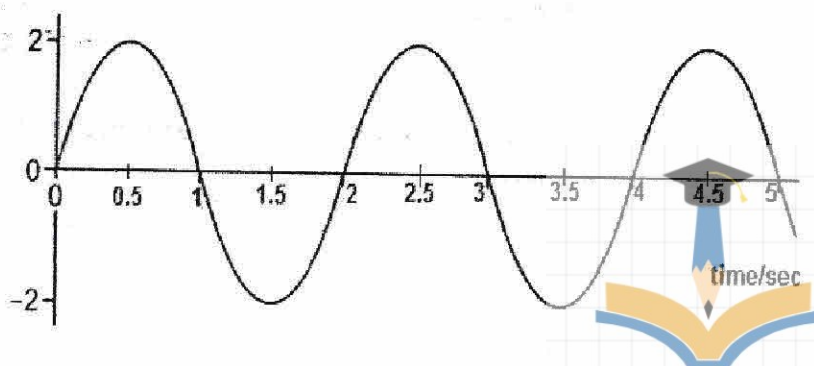
When the potential difference is 4V, the current is

- A. 0.3A
- B. 0.2A
- C. 0.1A
- D. 0.4A

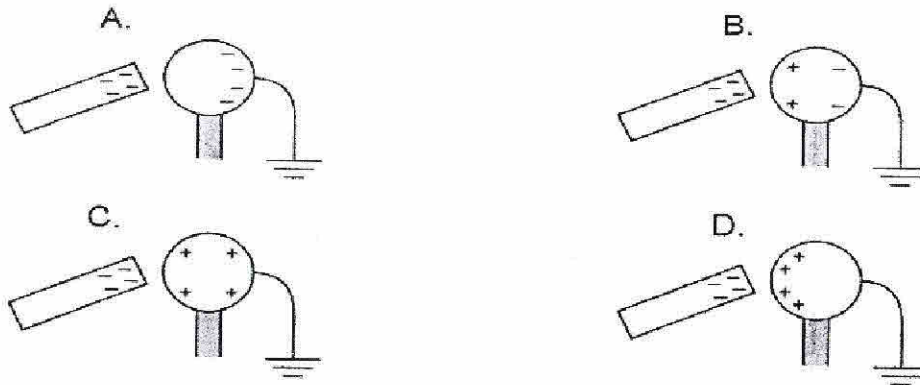
7. The diagram below shows a transverse wave.

The frequency of this wave is

- A. 0.5Hz
- B. 1 Hz
- C. 2 Hz
- D. 5 Hz



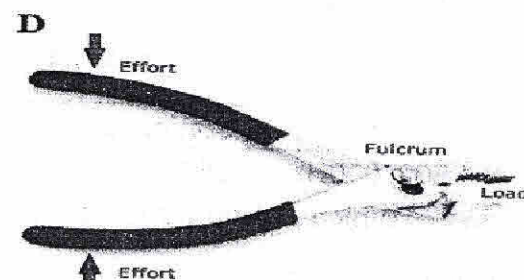
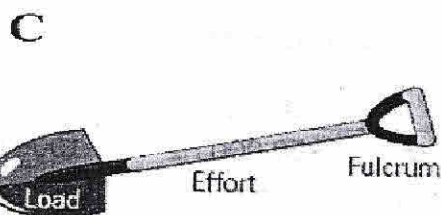
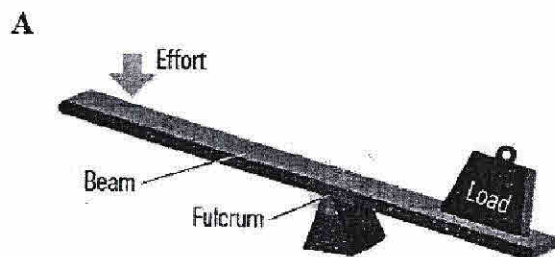
8. A negatively charged rod is held close to one side of a metal sphere with an insulating stand. The other side of the sphere is then earthed. Which diagram shows the charge distribution on the metal sphere?



9. The energy that the Earth receives from the sun is called

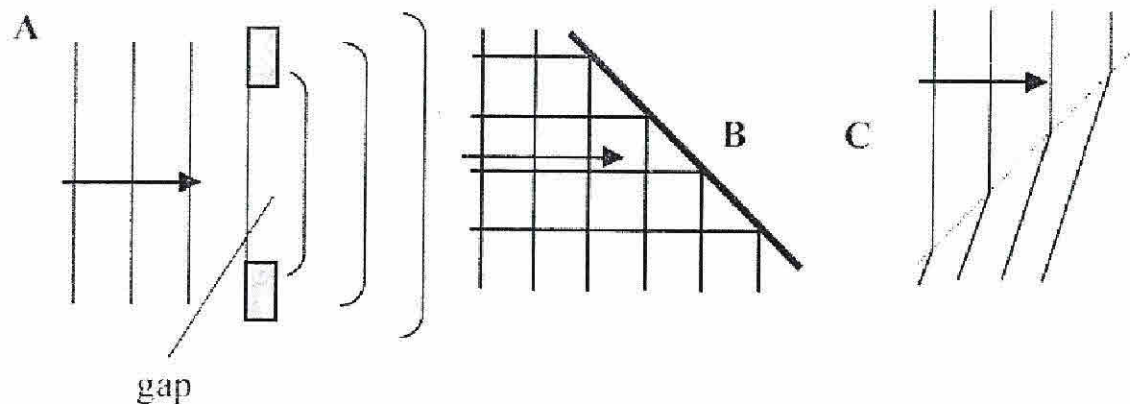
A. solar energy
B. geothermal energy
C. wind energy
D. hydroelectric energy

10. Which of the following simple machines is a third class of lever?



PART TWO: STRUCTURED QUESTIONS**(90 MARKS)****QUESTION ONE: WAVES (9 MARKS)**

A. Diagrams A, B and C show the behaviour of waves.



diffraction	interference	reflection
refraction	wavefront	wavelength

Us

e the word in the box above to complete the following

A is..... (1)

B is..... (1)

C is..... (1)

The diagram shows the types of radiation in the electromagnetic spectrum.

γ -rays	A	ultra-violet	Visible light	B	radio waves
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B. Name radiation A and radiation B.

A: (1)

B: (1)

C. Five radio stations transmit waves with the following frequencies

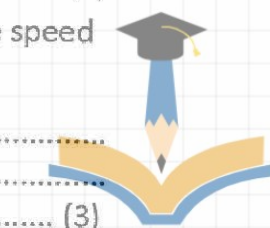
Station 1	Station 2	Station 3	Station 4	Station 5
500, 000Hz	550,000 Hz	600,000 Hz	850,000 Hz	1000,000 Hz

I. Which station transmits waves of the smallest wavelength?

..... (1)

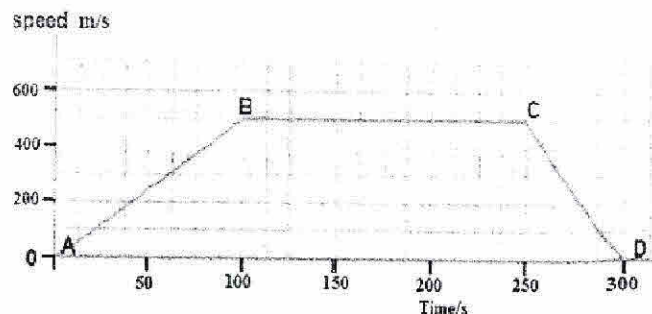
II. Find the wavelength of the waves whose frequency is 600,000Hz, if the speed of the radio waves in air is $3 \times 10^8 \text{ m/s}$.

..... (3)



QUESTION TWO: MOTION (12 marks)

- A. Fadumo cycles from her house to the supermarket each day. On a particular day her speed changes in the way shown on the graph.

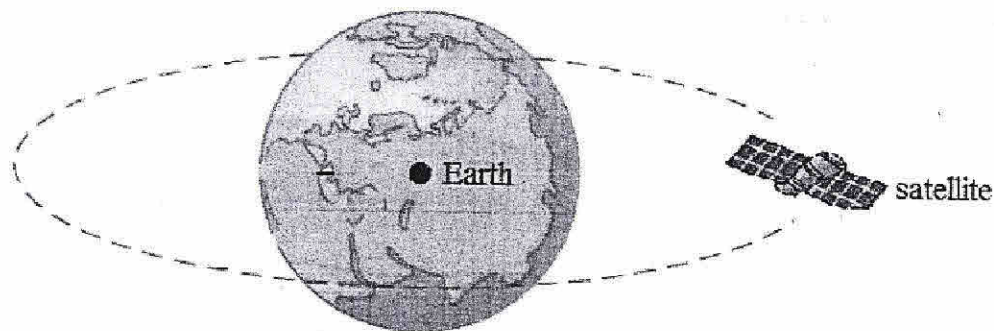


- The graph shows that Fadumo accelerates uniformly during the first seconds and decelerates uniformly during the lastseconds (2)
- Fadumo's speed along BC is m/s (1)
- Calculate the distance that Fadumo travelled in the first 100s

.....

 (3)

- A. The diagram below shows a geostationary satellite orbiting around the Earth.



What is the force that provides the centripetal force needed to orbit the satellite around the earth?

- (1)
- Is a larger or a smaller centripetal force required if
 - The satellite travels faster
 - The radius of the orbit is decreased
- (1)
- D. The satellite moves round a circular path of radius 6000m with a linear speed of 600m/s. Calculate centripetal acceleration

.....

 (3)

QUESTION THREE: FORCES (11MARKS)

A. By using the words in the box complete the blank spaces.

Impulse, momentum, moment, mass, inertia

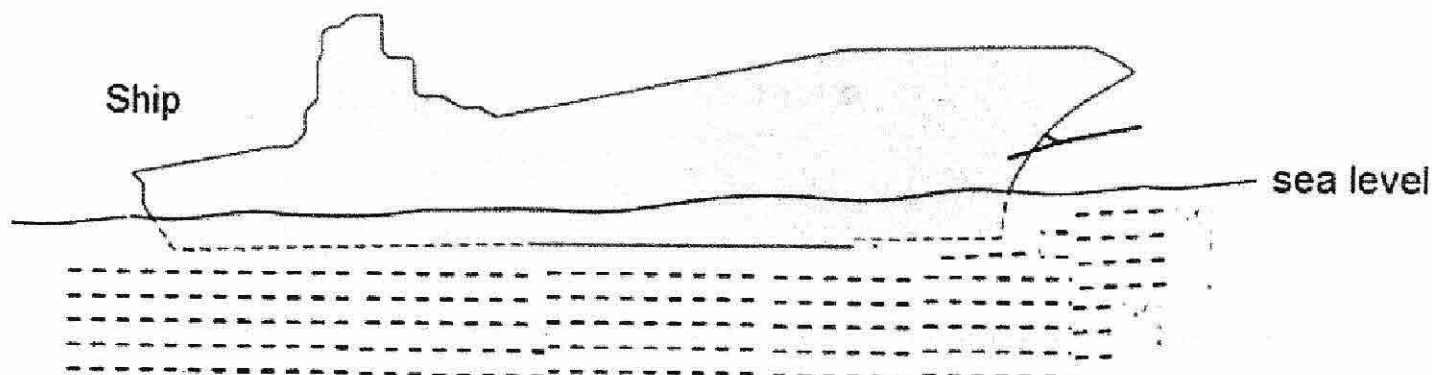
- II.Is the product of its velocity and mass (1)
- III.is The product of force and perpendicular distance from the line of action (1)
- IV. Is the resistance an object has to change its state of motion (1)

B. Calculate the momentum of a rugby player of mass 120 kg moving at 3m/s?

.....

 (3)

C. A ship is taken out to sea before being intentionally sunk. The ship is floating on the water and is not moving.



- D. On the diagram, use labeled arrow to show the direction of the buoyancy (upthrust) force acting on the ship. (1)
- E. State Archimedes's principle for floating and sinking?

.....

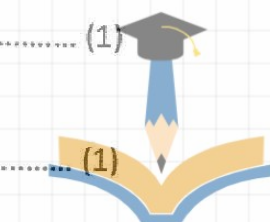
 (2)

F. If the weight of the ship is 10, 000N what is the buoyancy force of the water acting on the ship?

..... (1)

G. Explain why does a ship float but small steel nails sink?

..... (1)



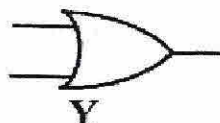
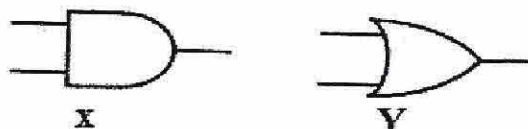
QUESTION FOUR: ELECTRONICS (9 MARKS)

Thermistor	Microphone	Loudspeaker
Light – Emitting Diode	Light Dependent Resistor	OR gate

A. Which one of the electronic devices above

- I. Is an output device which produces a sound _____ (1)
- II. Glows when a small current flows in it _____ (1)
- III. Has a lower resistance when it is heated _____ (1)
- IV. Has a lower resistance when light shines on it _____ (1)

B. The diagrams X and Y below show the symbols for two logic gates.



C. Which symbol represents an OR gate?

D. Complete the truth table for an OR gate

input 1	input 2	output
0	0	
0	1	
1	0	
1	1	

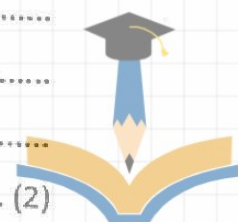
E. Highlight the importance of electronics to the real life using relevant examples from the local context.

.....

.....

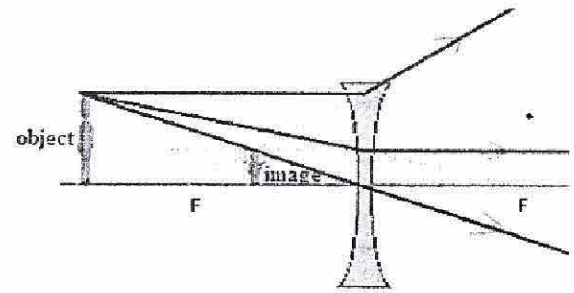
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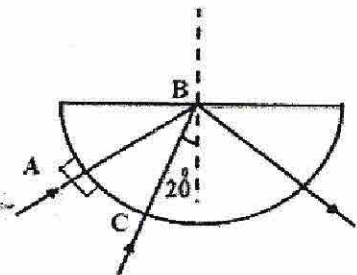
QUESTION FIVE: LIGHT (10 MARKS)

- A. The diagram shows an object and its image formed by a lens

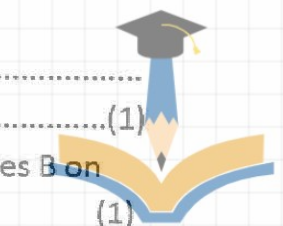


- I. State what type of lens convex or concave shown on the diagram?
..... (1)
 - II. Is the image real or virtual?
..... (1)
 - III. An object is placed 15cm from a convex lens of focal length 10 cm. Find the position of the image
.....
.....
.....
..... (2)
- B. State two practical applications of lenses
.....
..... (2)
- C. What color is formed when two primary colors Red and Green are overlapped?
..... (1)

- D. The diagram below shows rays of light in a semicircular glass, the critical angle of the glass is 42° .

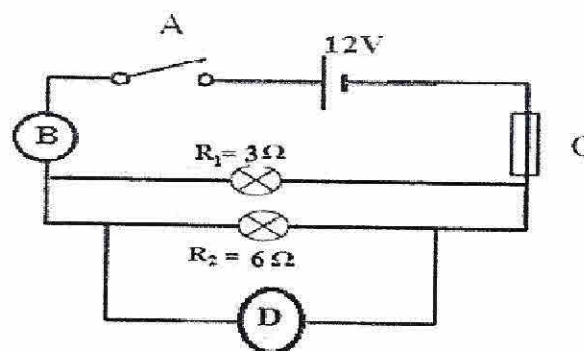


- i. Explain why the ray entering the glass at A is NOT bent.
..... (1)
- ii. Explain why the ray AB is reflected at B and NOT refracted.
..... (1)
- iii. Ray CB does NOT stop at B. Draw its approximate path after it leaves B on the diagram.
..... (1)



QUESTION SIX: ELECTRICITY (13 MARKS)

Warsame was carrying out an experiment in Bosaso Public school Physics laboratory. He drew a circuit diagram using symbols to represent the components he used to build his circuit as shown right.



A. Complete the blanks using appropriate components

- I. A is _____ (1)
- II. B is _____ measuring the total current flowing both bulbs (1)
- III. D is _____ measuring the potential difference across the lower bulb (1)

B. Calculate:

- I. The total resistance R_1 and R_2 of the two bulbs

.....

(2)

- II. The total current flowing through both bulbs when A is closed

.....

(2)

- IV. The component C is a circuit symbol of a fuse. Explain how the fuse prevents too large a current in the circuit.

.....

(2)

- III. In mains circuit the fuse is placed in the **live wire** instead of the **neutral wire** why?

.....
(1)

- IV. If energy costs 10 s.sh per unit, calculate the cost of using 3kW electric fire for 5hrs

.....

(3)



QUESTION SEVEN: RADIOACTIVITY (9MARKS)

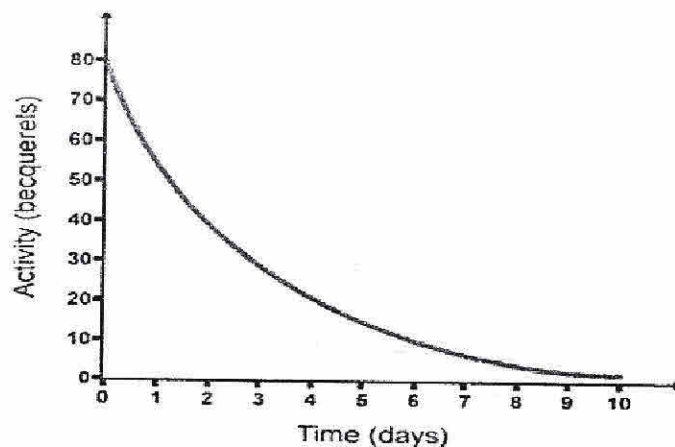
A. The graph shows how the activity of a small radioactive sample varies with time.

i. From the graph estimate the half – life of the sample

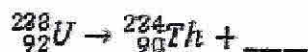
..... (1)

ii. What would be the activity of the sample after 6 days?

..... (1)



B. When radioactive Uranium –238 decays, Thorium –234 is formed.

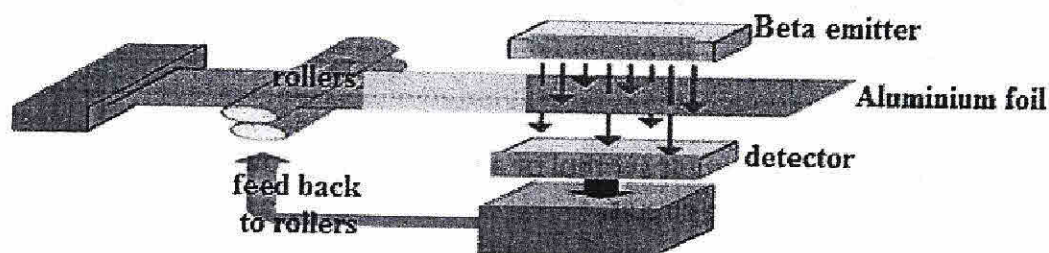


Assuming that only one particle is emitted.

i. What type of particle is emitted? (1)

ii. What is the relative charge of this particle? (1)

C. The diagram below shows thickness monitoring system of Beta radiation.



I. Explain why beta is used rather than Alpha or Gamma emitter?

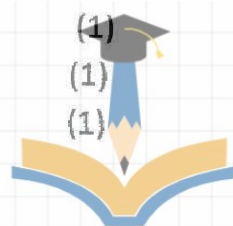
.....

 (2)

D. Radiations emitted from nucleus of unstable atoms are Alpha, Beta and Gamma.

Which type of radiations

- I. Is the least penetrating.....?
- II. Is not deflected by an electric field.....?
- III. Has a negative charge.....?



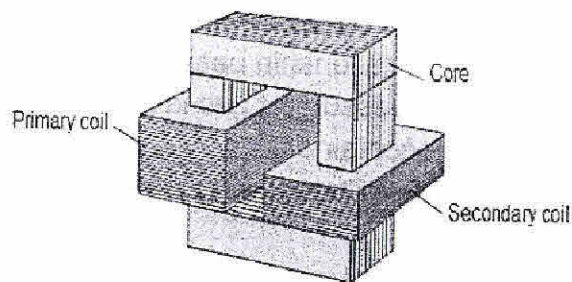
QUESTION EIGHT: ELECTROMAGNETIC INDUCTION (8 MARKS)

A. A teacher demonstrates a small transformer

Which part of the transformer is made of

i. Iron (1)

ii. Copper (1)



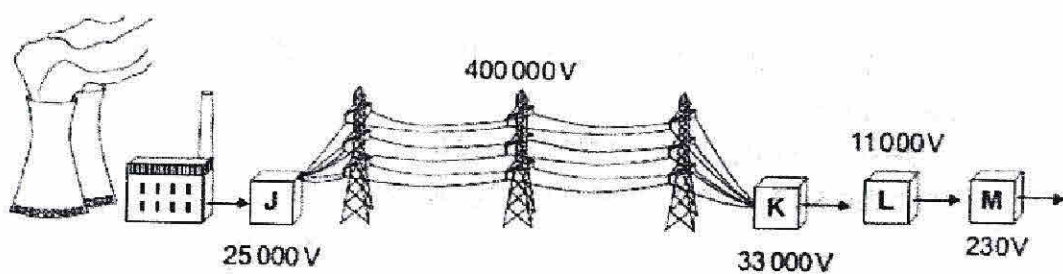
B. Suggest a reason why the core is used in the transformer

..... (1)

C. The number of turns in the secondary coil is less than the number of turns in the primary coil. What sort of transformer is it?

..... (1)

D. The diagram shows part of the National Grid system. The transformers, J, K, L and M, are an essential part of the system.



i. Which transformer, J, K, L or M, is a step-up transformer?

..... (1)

ii. Why a step-up transformer used as part of the National Grid?

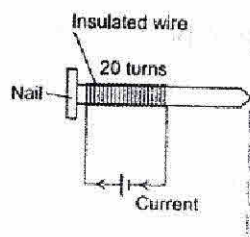
..... (1)

E. Explain why the number of clips attached on nail A are more than number of clips on Nail B?

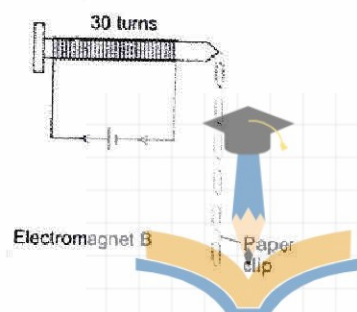
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.....

..... (2)



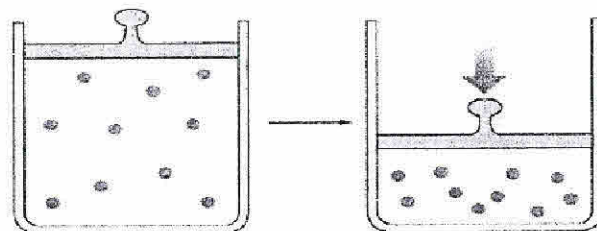
Electromagnet A



Electromagnet B

QUESTION NINE: HEAT & TEMPERATURE (9 MARKS)

- A. The diagram below shows the relationship between volume and pressure of gas inside enclosed container.



- I. State the **Boyle's law** of gases?

.....

(1)

- II. Define pressure(1)

- III. 20 cm^3 of gas has a pressure of 15 pa. If the pressure is reduced to 5 pa, calculate new volume of the gas?

.....

(2)

- B. The diagram shows a solar heating panel which is used to heat cold water in a house.

a. Describe how heat is transferred

- i. From the sun to the solar heating panel

.....
(1)

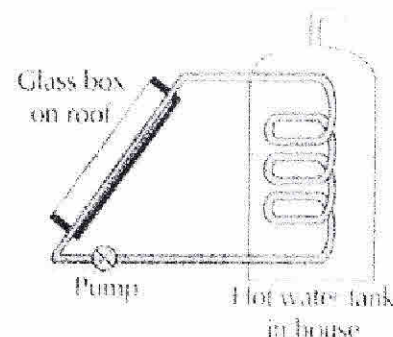
- ii. From the hot water in the pipe to the cold water in the tank.

.....(1)

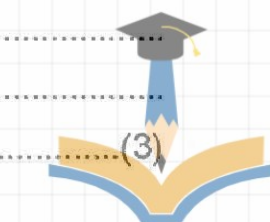
- iii. Calculate the quantity of heat absorbed by the water in the tank if the mass of water is 20 kg and temperature rises from 20°C to 50°C given that of specific heat capacity of water is $4200 \text{ J/kg}^\circ\text{C}$.

.....

(3)



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